

WEEK 4 – ASSIGNMENT

Name- Hamed Raza
Amity University
A023119823002

Title: Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

The screenshot shows the 'Create a resource group' page in the Azure Portal. The browser tabs include 'LMS', 'Superstore Dataset', 'Mail - HAMED RAZA - Outlook', and 'Create a resource group - Micr...'. The address bar shows the URL: 'portal.azure.com/?Microsoft_Azure_Education_correlationId=2f3d444e-94fc-4240-a282-e8a21e06595f8&Microsoft_Azure_Education_newA4E=true&Microsoft_Azure_Educati...'. The page header includes 'Microsoft Azure' and a search bar. The breadcrumb trail is 'Home > Resource Manager | Resource groups'. The main heading is 'Create a resource group'. Below this, there are tabs for 'Basics', 'Tags', and 'Review + create'. The 'Basics' tab is active. A description of a resource group is provided: 'Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. Learn more'. Three form fields are present: 'Subscription *' with a dropdown menu showing 'Azure for Students', 'Resource group name *' with a text input field containing 'Resource-gr', and 'Region *' with a dropdown menu showing '(Asia Pacific) Central India'. At the bottom, there are three buttons: 'Previous', 'Next', and 'Review + create'.

Create a resource group ...

Basics

Tags

Review + create

[Automation Link](#)

Basics

Subscription

Azure for Students

Resource group name

Resource-gr

Region

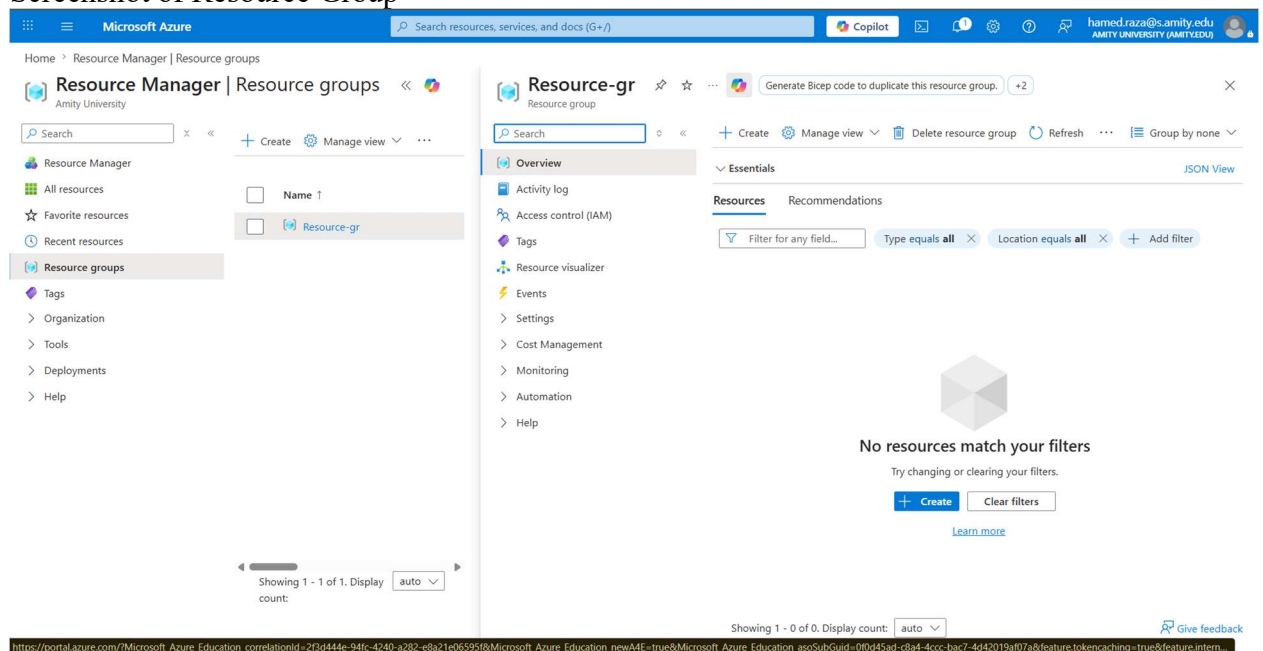
Central India

Tags

None

Deliverable:

- Screenshot of Resource Group



Task 2: Storage Setup

Do:

- Create a Storage Account

The screenshot shows the Azure portal interface for a Storage Account named 'ceistorage2026'. The left sidebar contains a navigation menu with options like Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Storage Mover, Partner solutions, Resource visualizer, Data storage, Security + networking, Data management, Settings, Monitoring, Automation, and Help. The main content area is divided into two sections: 'Essentials' and 'Properties'. The 'Essentials' section displays key information about the storage account, including its Resource group (Resource-gr), Location (centralindia), Subscription (Azure for Students), Subscription ID (0f0d45ad-c8a4-4ccc-bac7-4d42019af07a), Disk state (Available), and Tags (Add tags). The 'Properties' section is further divided into 'Blob service' and 'Security' tabs. The 'Blob service' tab shows settings for Hierarchical namespace (Disabled), Default access tier (Hot), Blob anonymous access (Disabled), Blob soft delete (Enabled (7 days)), Container soft delete (Enabled (7 days)), Versioning (Disabled), Change feed (Disabled), NFS v3 (Disabled), Allow cross-tenant replication (Disabled), and Storage tasks assignments (None). The 'Security' tab shows settings for Require secure transfer for REST API operations (Enabled), Storage account key access (Enabled), Minimum TLS version (Version 1.2), Infrastructure encryption (Disabled), Public network access (Enabled), Public network access scope (Enable from all networks), Private endpoint connections (0), and Network routing (Microsoft network routing).

- Create a Blob Container

The screenshot shows the Azure portal interface for a Storage Account named 'ceistorage2026', specifically the 'Containers' page. The left sidebar contains a navigation menu with options like Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Storage Mover, Partner solutions, Resource visualizer, Data storage, Containers, Classic file shares, Queues, Tables, Security + networking, Data management, Settings, and Help. The main content area displays a table of containers. The table has columns for Name, Last modified, Anonymous access level, Lease state, and Storage connector. There are two containers listed: 'slogs' and 'input-data'. Both containers have a Private Anonymous access level and an Available Lease state. The 'input-data' container was last modified on 5/31/2026 at 1:28:30 PM.

- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file

input-data

Search

+ Add Directory

↑ Upload

🔗 Change access level

🔄 Refresh

🗑️ Delete

📄 Copy

📄 Paste

🔄 Rename

🔗 Acquire lease

🔗 Break lease

⚙️ Edit columns

Overview

- 🔧 Diagnose and solve problems
- 🔑 Access Control (IAM)
- > Settings

input-data

Authentication method: Access key [\(Switch to Microsoft Entra user account\)](#)

🔍 Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	📄 Sample - Superstore.csv	5/31/2026, 1:29:36 PM	Hot (Inferred)	Block blob	2.18 MiB	Available	⋮

WEEK 4 - ASSIGNMENT

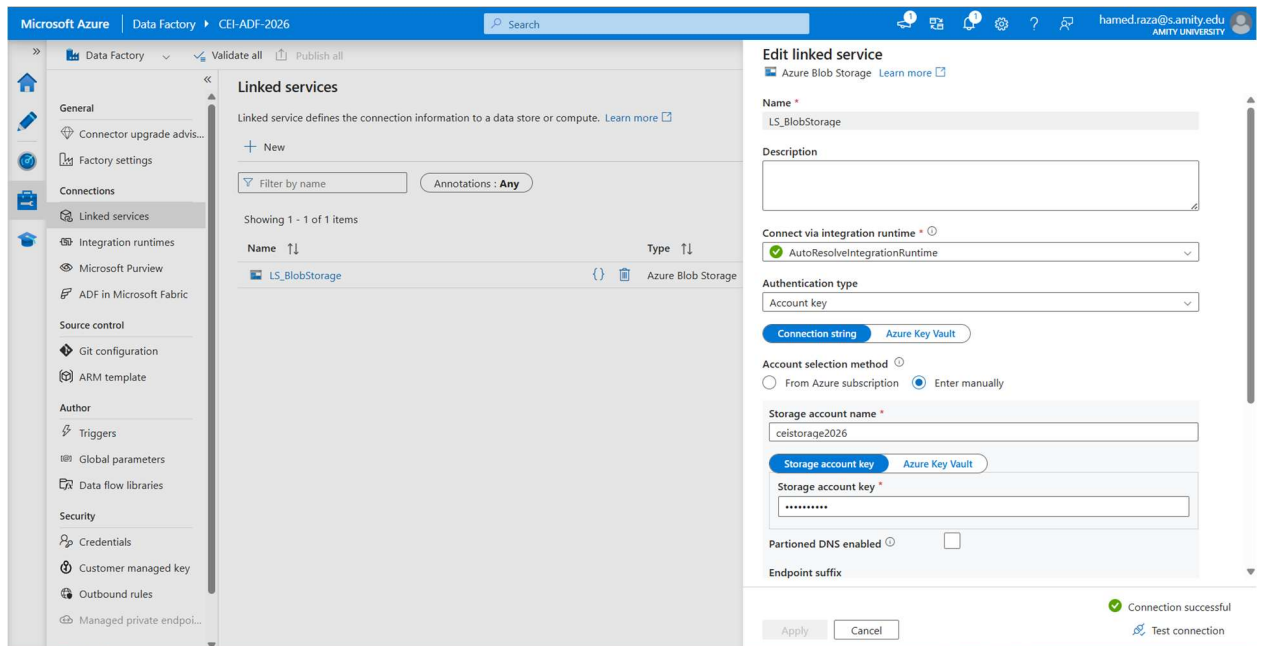
Task 3: ADF Basics

Do:

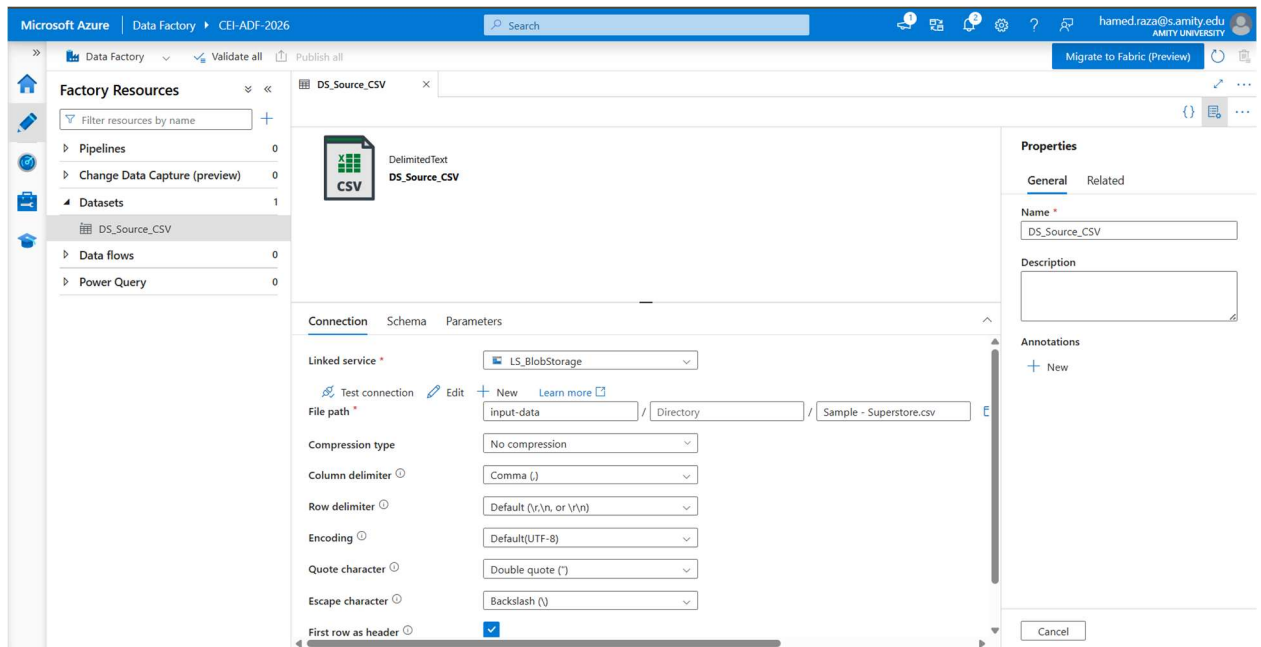
- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

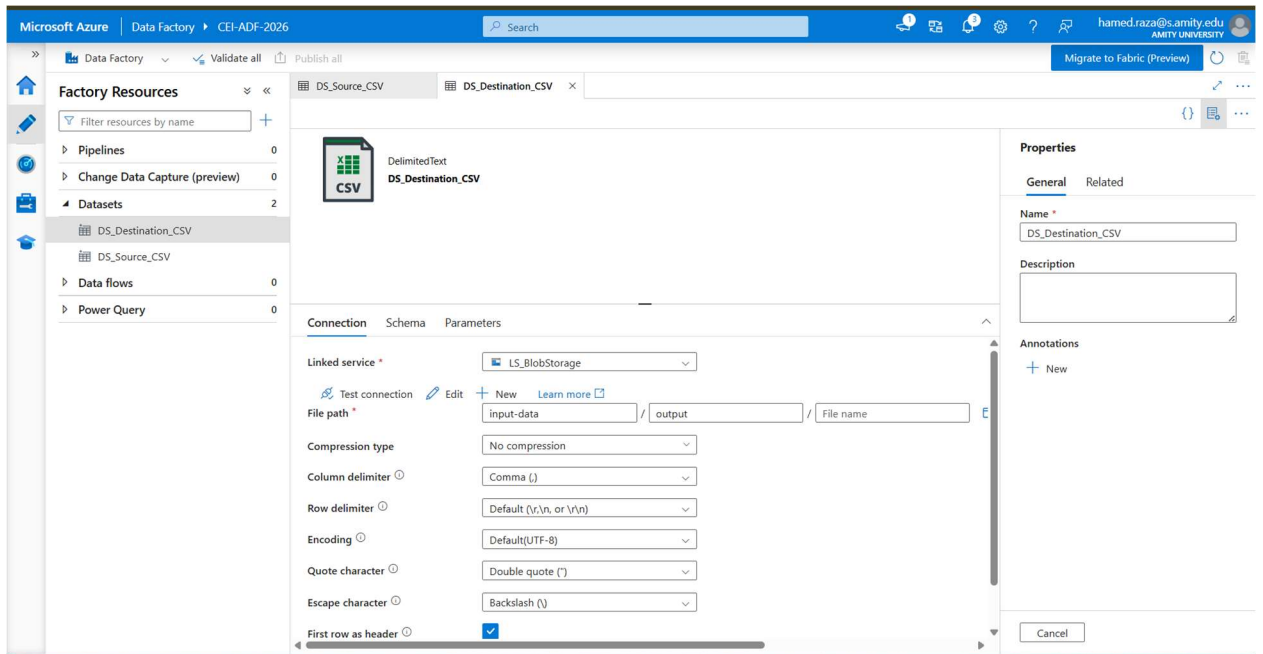
Deliverable:

- Screenshot of Linked Service

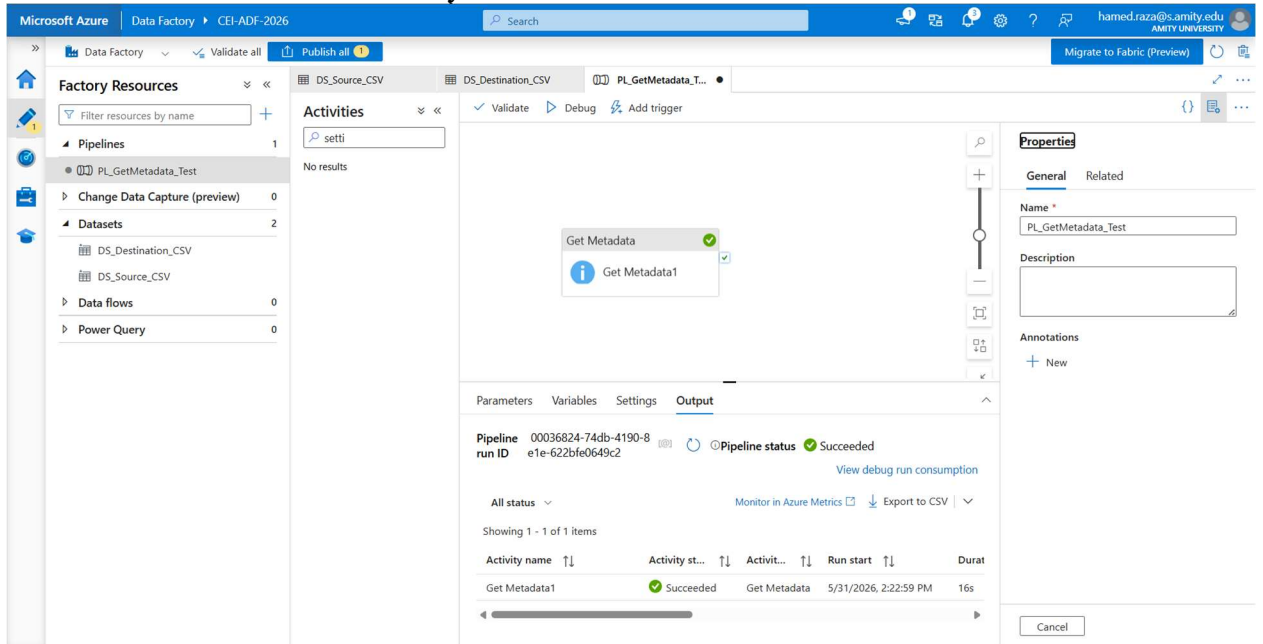


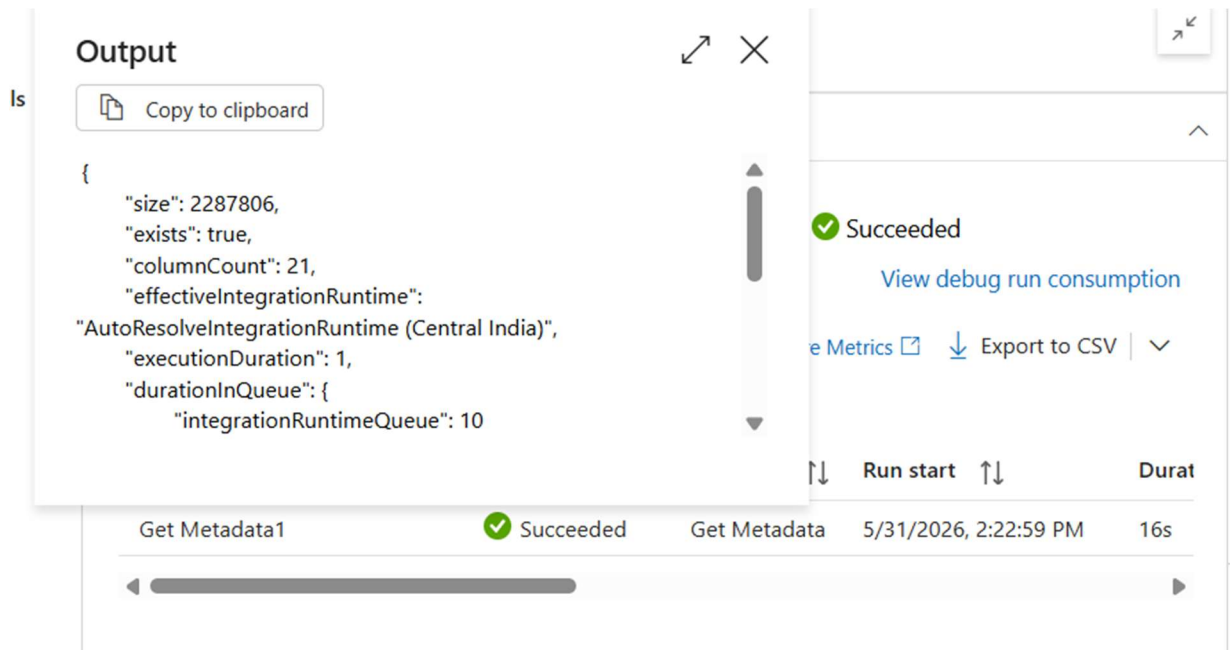
- Screenshot of dataset





- Screenshot of Get Metadata activity





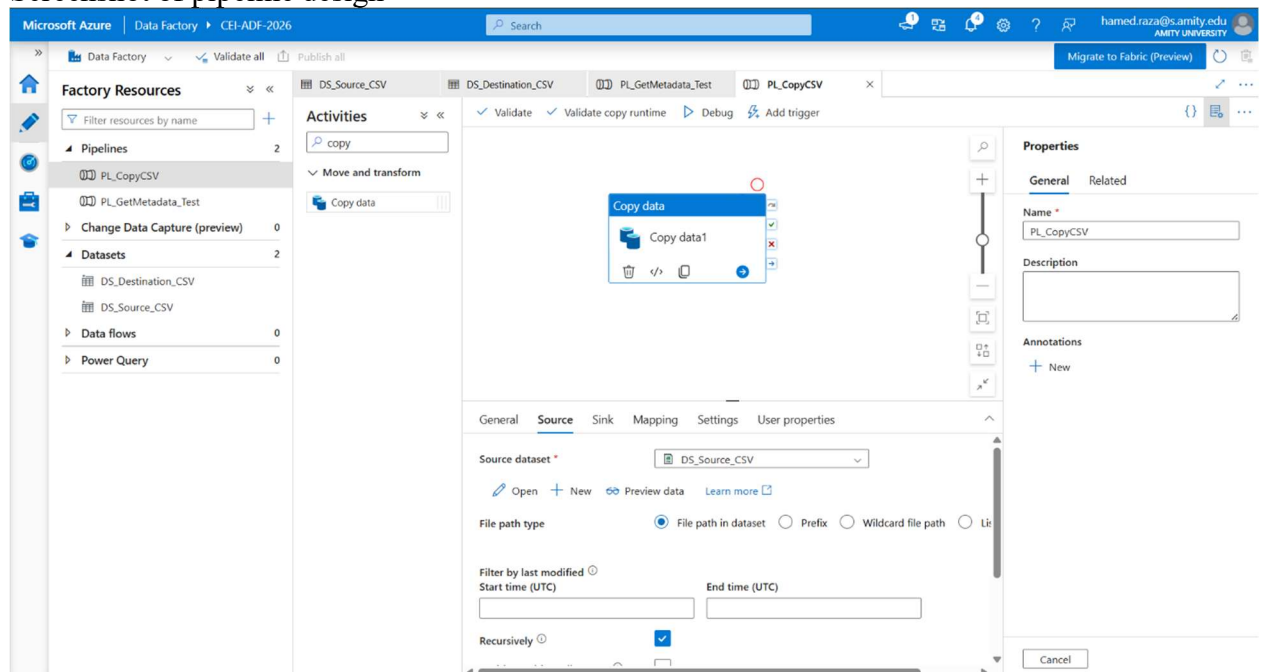
Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design



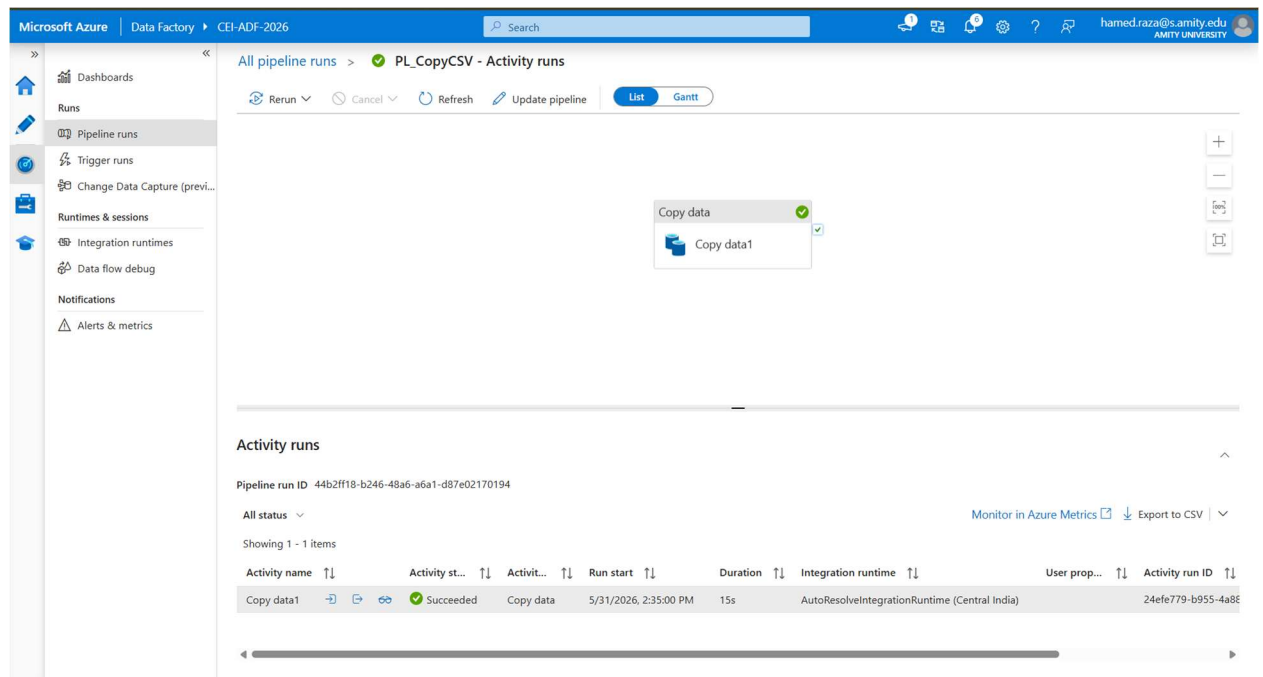
Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)



Microsoft Azure | Data Factory | CEI-ADF-2026

All pipeline runs > PL_CopyCSV - Activity runs

Renun Cancel Refresh Update pipeline List Gantt

Copy data

Copy data1

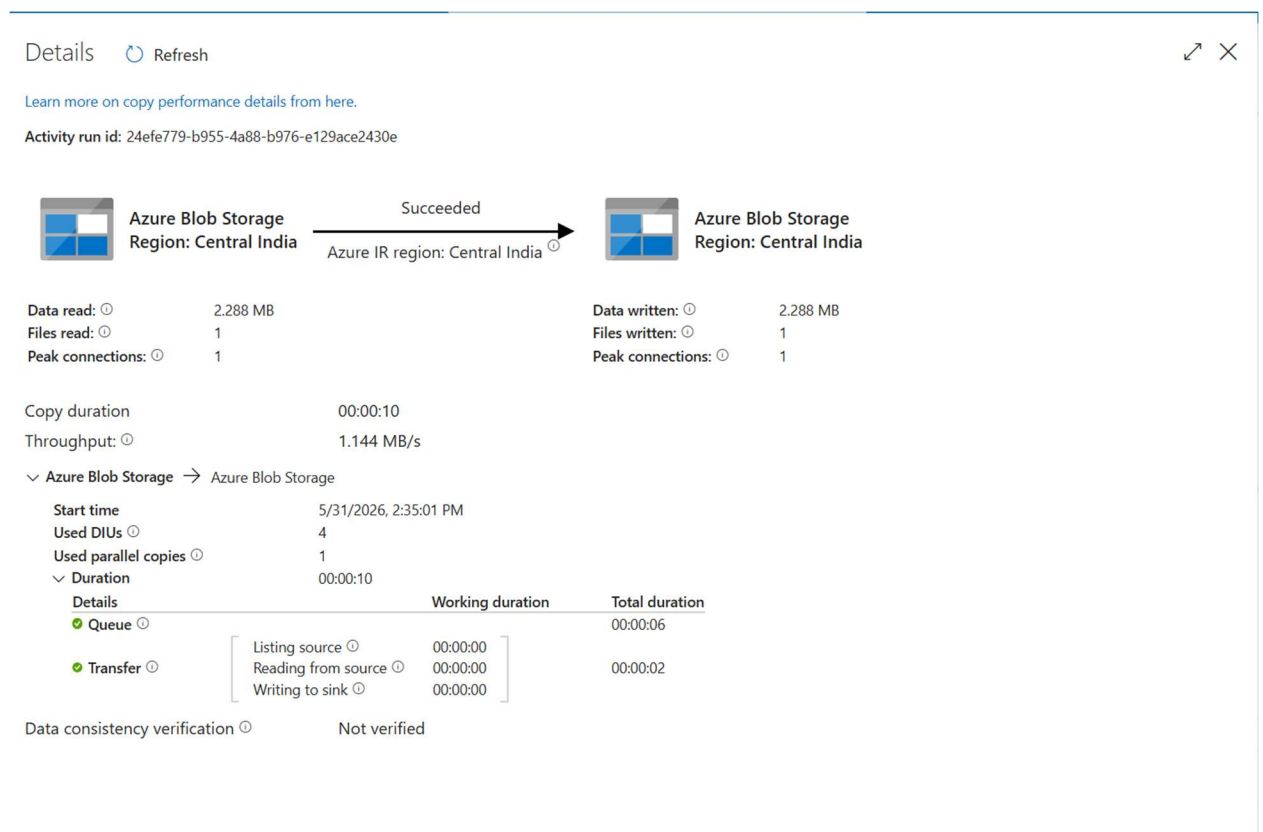
Activity runs

Pipeline run ID: 44b2ff18-b246-48a6-a6a1-d87e02170194

All status

Showing 1 - 1 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
Copy data1	Succeeded	Copy data	5/31/2026, 2:35:00 PM	15s	AutoResolveIntegrationRuntime (Central India)		24efe779-b955-4a88



Details Refresh

Learn more on copy performance details from here.

Activity run id: 24efe779-b955-4a88-b976-e129ace2430e

Azure Blob Storage Region: Central India → Succeeded → Azure Blob Storage Region: Central India

Azure IR region: Central India

Data read: 2.288 MB
Files read: 1
Peak connections: 1

Data written: 2.288 MB
Files written: 1
Peak connections: 1

Copy duration: 00:00:10
Throughput: 1.144 MB/s

✓ Azure Blob Storage → Azure Blob Storage

Start time: 5/31/2026, 2:35:01 PM
Used DIs: 4
Used parallel copies: 1

✓ Duration: 00:00:10

Details	Working duration	Total duration
Queue		00:00:06
Transfer		00:00:02
Listing source	00:00:00	
Reading from source	00:00:00	
Writing to sink	00:00:00	

Data consistency verification: Not verified

input-data ...

Container

- + Add Directory
- ↑ Upload
- ↻ Refresh
- 🗑 Delete
- 📄 Copy
- 📄 Paste
- 🔄 Rename
- 🔑 Acquire lease
- 🔑 Break lease
- 📄 Edit columns

Overview

🔧 Diagnose and solve problems

👤 Access Control (IAM)

> Settings

input-data > output

Authentication method: Access key (Switch to Microsoft Entra user account)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	[-]						...
☒	Sample - Superstore.csv	5/31/2026, 2:35:11 PM	Hot (Inferred)	Block blob	2.18 MiB	Available	...

WEEK 4 - ASSIGNMENT

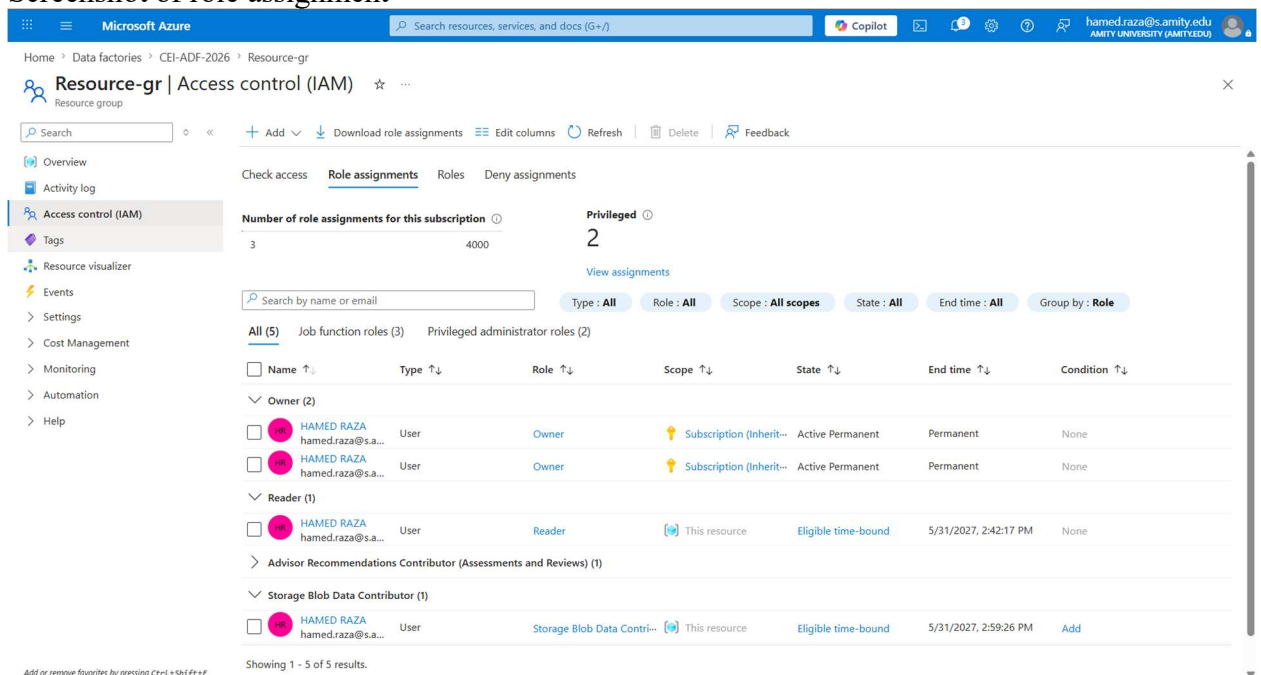
Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment



Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully
- Data copied to destination
- Metadata validated

Microsoft Azure | Data Factory | CEI-ADF-2026

All pipeline runs > PL_CopyCSV - Activity runs

Rerun Cancel Refresh Update pipeline List Gantt

Get Metadata Copy data

Get Metadata1 Copy data1

Activity runs

Pipeline run ID: 4c455906-8d61-420b-a8a2-488cc714a58f

All status Monitor in Azure Metrics Export to CSV

Showing 1 - 2 items

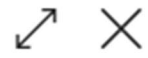
Activity name	Activity status	Activity name	Run start	Duration	Integration runtime	User properties	Activity run ID
Copy data1	Succeeded	Copy data	5/31/2026, 3:05:17 PM	15s	AutoResolveIntegrationRuntime (Central India)		29badc2e-2798-4111
Get Metadata1	Succeeded	Get Metadata	5/31/2026, 3:05:03 PM	13s	AutoResolveIntegrationRuntime (Central India)		8d5057e9-48a6-4db

Output

Copy to clipboard

```
{
  "exists": true,
  "size": 2287806,
  "columnCount": 21,
  "effectiveIntegrationRuntime":
    "AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 1,
  "durationInQueue": {
    "integrationRuntimeQueue": 7
  }
}
```

Output



Copy to clipboard

[Learn more on output details](#)

```
{  
  "dataRead": 2287806,  
  "dataWritten": 2287806,  
  "filesRead": 1,  
  "filesWritten": 1,  
  "sourcePeakConnections": 1,  
  "sinkPeakConnections": 1,  
  "copyDuration": 11,  
  "throughput": 1143.903,  
}
```

